

IT1X24 – Statistical Research Methods

Part 1: Hypothesis Testing Report

A. Introduction

1. Context of the Social Issue

In recent years, social media usage among adolescents and young adults has increased significantly due to widespread smartphone access and digital connectivity. Platforms such as Instagram, TikTok, Snapchat, and YouTube are widely used for communication, entertainment, and content consumption. However, prolonged exposure to social media—particularly during nighttime—has raised concerns regarding its potential impact on sleep duration.

Sleep is essential for physical health, cognitive functioning, emotional regulation, and academic performance. Insufficient sleep has been associated with decreased concentration, increased stress levels, mood instability, and long-term health risks. Health authorities generally recommend that young adults obtain at least 7 hours of sleep per night.

Polytechnic students often juggle academic workload, social commitments, and part-time employment, which may contribute to irregular sleep patterns. Furthermore, late-night screen usage and “doomscrolling” behaviours may delay sleep onset and reduce total sleep duration.

This study adopts a **quantitative experimental research design** to examine sleep duration among polytechnic students and to determine whether reducing social media screen time leads to an increase in sleep duration.

2. Research Statements

This study investigates the relationship between social media usage and sleep duration through two hypothesis tests:

One-Sample t-test Research Statement

Polytechnic students sleep fewer hours per night than the recommended national average of 7 hours.

Paired t-test Research Statement

Reducing social media screen time increases daily sleep duration among polytechnic students.

3. Hypotheses

A. One-Sample t-test

Let μ represent the mean sleep duration of polytechnic students.

Null Hypothesis (H_0):

$$\mu \geq 7$$

Alternative Hypothesis (H_1):

$$\mu < 7$$

This is a **left-tailed test**, conducted at $\alpha = 0.05$.

B. Paired t-test

Let μ_d represent the mean difference in sleep duration (After reduction – Before reduction).

Null Hypothesis (H_0):

$$\mu_d \leq 0$$

Alternative Hypothesis (H_1):

$$\mu_d > 0$$

This is a **right-tailed test**, conducted at $\alpha = 0.05$.

B. Methods

4. Data Collection

Target Population

The target population consists of polytechnic students in Singapore.

Sampling Method

A convenience sampling method was used. Participants were recruited through online distribution of a survey form.

Sample Size

A total of **40 valid responses** were collected ($n = 40$).

Variables Collected

For the one-sample t-test:

- Average sleep hours on a typical weekday (continuous variable)

For the paired t-test:

- Sleep hours before reducing social media usage
- Sleep hours after reducing social media usage

All responses were measured in hours.

Data Quality Measures

- Only polytechnic students were included.
- Sleep duration was recorded numerically in hours.
- The same respondents provided both "before" and "after" measurements for the paired analysis.
- Units were standardised in hours to ensure consistency.

5. Hypothesis Testing Plan

Two statistical tests were conducted:

1. One-sample t-test to compare the sample mean sleep duration with the benchmark value of 7 hours.
2. Paired t-test to evaluate within-person changes in sleep duration after reducing screen time.

Level of significance:

$$\alpha = 0.05$$

Rejection rule:

- For the one-sample left-tailed test, reject H_0 if $t < -1.685$ ($df = 39$).
- For the paired right-tailed test, reject H_0 if $t > 1.685$ ($df = 39$).

All calculations were performed using standard t-test formulas.

C. Results

6.1 Descriptive Statistics

For average sleep duration (typical weekday):

- Sample size (n) = 40
- Mean sleep duration ($\bar{x}$) = **5.6 hours**
- Standard deviation (s) = **1.03 hours**

The average sleep duration is below the recommended 7 hours, suggesting potential sleep deprivation among respondents.

6.2 One-Sample t-test

Test statistic formula:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substituting values:

- $\bar{x} = 5.6$
- $\mu_0 = 7$
- $s = 1.03$
- $n = 40$

Computed test statistic:

$t = \mathbf{-8.57}$

Degrees of freedom (df) = 39

Critical value ($\alpha = 0.05$, left-tailed, $df=39$):

$t_{critical} = -1.685$

Since $-8.57 < -1.685$, the test statistic falls within the rejection region.

Decision

Reject H_0 .

Interpretation

At the 5% significance level, there is sufficient statistical evidence to conclude that polytechnic students sleep significantly fewer than 7 hours per night on average.

6.3 Paired t-test

Mean difference (After – Before) = **+1.5 hours**

Test statistic formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Computed test statistic:

$t = 18.73$

Degrees of freedom (df) = 39

Critical value ($\alpha = 0.05$, right-tailed, df=39):

$t_{\text{critical}} = 1.685$

Since $18.73 > 1.685$, the test statistic falls within the rejection region.

Decision

Reject H_0 .

Interpretation

There is strong statistical evidence that reducing social media screen time significantly increases daily sleep duration among polytechnic students.

The observed increase of 1.5 hours suggests a substantial improvement in sleep duration following reduced screen time.

D. Discussion and Conclusion

7. Discussion

The findings indicate that polytechnic students sleep significantly fewer than the recommended 7 hours per night. This supports growing concerns about sleep deprivation among young adults.

Furthermore, the paired t-test demonstrates that reducing social media usage results in a statistically significant increase in sleep duration. The average increase of 1.5 hours suggests that screen time plays a substantial role in influencing sleep patterns.

One possible explanation is that social media exposure before bedtime delays sleep onset due to cognitive stimulation and blue light exposure, which may suppress melatonin production.

Implications

- Students should consider limiting social media usage before bedtime.
- Educational institutions may promote digital wellness initiatives.
- Awareness campaigns can encourage healthier sleep habits.

Limitations

- Data was self-reported and may be subject to response bias.
- Convenience sampling limits generalisability.
- The intervention duration was relatively short.
- Other confounding factors (e.g., caffeine intake, academic stress) were not experimentally controlled.

Recommendations for Future Research

Future studies could:

- Use objective sleep tracking devices.
- Apply random sampling methods.
- Conduct longer intervention periods.
- Control for confounding variables.

Conclusion

This study provides strong statistical evidence that polytechnic students experience sleep deprivation and that reducing social media screen time significantly improves sleep duration. The findings highlight the importance of digital habit management in promoting student well-being.